

# Cheol Hyun Park

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## EDUCATION

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**Ph.D. Materials Science and Engineering (MSE),**  
**Sungkyunkwan University (SKKU),** Graduate School  
Advisor: Prof. Jung Heon Lee (email: jhlee7@skku.edu)

Mar. 2020–Aug. 2026  
Suwon, South Korea

**B.S. MSE, SKKU**

Mar. 2014–Feb. 2020  
Suwon, South Korea

## RESEARCH INTERESTS

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Nanoparticle, Self-assembly, Ceramics, Biomaterials

## RESEARCH FOCUS

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**Advanced Characterization via Electron Microscopy:** Research focuses on elucidating the organic-inorganic hybrid interface architecture of bioinspired ceramic composites using high-resolution electron microscopy. Expertise includes SEM-EDS mapping to visualize morphological and elemental distributions in hierarchical ceramic structures, and TEM analysis to investigate nanoscale interfacial features and defect structures. These structural insights are directly correlated with multi-scale mechanical behavior, from nanoscale (AFM), through microscale (nanoindentation), to macroscale (UTM, fracture toughness test), to elucidate structure-property-performance relationships in advanced ceramics.

## PUBLICATIONS

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### First and Co-first author publications (<sup>†</sup>*Equal contribution*)

- [7] **C. H. Park**<sup>†</sup>, P. R. Sundharbaabu<sup>†</sup>, Yoon Sung Cho<sup>†</sup>, J. H. Heo\*, J. H. Kang\*, J. H. Lee\*, “Engineering "Plug-and-Play" Functional Metallized Ceramics Using DNA as Synthesis Scaffolds” (*To be submitted*)
- [6] J. Cho<sup>†</sup>, **C. H. Park**<sup>†</sup>, P. R. Sundharbaabu<sup>†</sup>, S.-U. Ju, D. Shin, J. H. Heo\*, M. Kim\*, J. H. Lee\*, “Transformative Mechanical and Piezoelectric Properties in Organic–Inorganic Hybrid Materials for Biomedical Applications” *ACS Nano* **19** (41), 36035–36081 (2025).
- [5] **C. H. Park**<sup>†</sup>, H. Song<sup>†</sup>, D.-H. Kim, J. Kim, S. H. Jeong, J. B. B. Rayappan, J. H. Heo, J. H. Lee, “Functional ATP-Based Fabric Organic Dyes with Flame Retardancy and Antibacterial Activity” *Chemical Engineering Journal* **506**, 159912 (2024).
- [4] J. W. Lee<sup>†</sup>, B. Lee<sup>†</sup>, **C. H. Park**<sup>†</sup>, J. H. Heo, T. Y. Lee, D. Lee, J. Bae, P. R. Sundharbaabu, W. K. Yeom, S. Chae, J.-H. Lim, S.-W. Lee, J.-S. Choi, H.-B. Bae, J.-Y. Choi, E.-H. Lee, D. S. Yoon, G. Y. Yeom, H. Shin, J. H. Lee, “Monolithic DNApatite: An Elastic Apatite with Sub-Nanometer Scale Organo–Inorganic Structures” *Advanced Materials* **36** (41), 2406179 (2024).
- [3] H. Song<sup>†</sup>, **C. H. Park**<sup>†</sup>, S. H. Jeong, J. H. Heo, J. H. Lee, “Synergistic adenosine triphosphate/chitosan bio-coatings on polyurethane foam for simultaneously improved flame retardancy and smoke suppression” *ACS applied Polymer Materials* **5** (6), 4388–4399 (2023).

- [2] S. H. Jeong<sup>†</sup>, **C. H. Park<sup>†</sup>**, H. Song, J. H. Heo, J. H. Lee, “Biomolecules as green flame retardants: recent progress, challenges, and opportunities” *Journal of Cleaner Production* **368**, 133241 (2022).
- [1] S. H. Kim<sup>†</sup>, **C. H. Park<sup>†</sup>**, J. H. Heo, J. H. Lee, “Progress and perspectives of metal-ion-substituted hydroxyapatite for bone tissue engineering: comparison with hydroxyapatite” *Journal of Korean Ceramic Society* **59** (3) (2022).

#### Co-author publications

- [4] H. Song<sup>†</sup>, S. H. Jeong<sup>†</sup>, **C. H. Park**, M. J. Kim, H. Kim, J. H. Heo, J. H. Lee, “A multifaceted heat-switching nanofiller: Design of heat dissipation–insulation switching materials above a critical temperature” *Chemical Engineering Journal* **497**, 154406 (2024).
- [3] D.-H. Kim<sup>†</sup>, J. Bae<sup>†</sup>, J. H. Heo, **C. H. Park**, E. B. Kim, J. H. Lee, “Nanoparticles as next-generation tooth-whitening agents: Progress and perspectives” *ACS Nano* **16** (7), 10042–10065 (2022).
- [2] C. Kim<sup>†</sup>, J. W. Lee<sup>†</sup>, J. H. Heo, **C. H. Park**, D.-H. Kim, G. S. Yi, H. C. Kang, H. S. Jung, H. Shin, J. H. Lee, “Natural bone-mimicking nanopore-incorporated hydroxyapatite scaffolds for enhanced bone tissue regeneration” *Biomaterials Research* **26** (1), 7 (2022).
- [1] S. H. Jeong<sup>†</sup>, J. H. Heo, J. W. Lee, M. J. Kim, **C. H. Park**, J. H. Lee, “Bioinspired adenosine triphosphate as an “all-in-one” green flame retardant via extremely intumescent char formation” *ACS Applied Materials & Interfaces* **13** (19), 22935–22945 (2021).

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#### CONFERENCE

#### Contributed Talks

- [6] **C. H. Park**, J. H. Lee\*, “Biomimetic Monolithic Ceramic-Polymer Nanohybrid with Sub-Nanometer Precision for Enhanced Mechanical Stability and Sustainable Support” Oral presentation, *International Conference on Nanogenerators and Piezotronics (NGPT 2026)*, Seoul, Korea. June 9–12.
- [5] **C. H. Park**, J. W. Lee, B. Lee, D. Lee, J. H. H, T.-Y. Lee, J. Bae, P. R. Sundharbaabu, J. H. Lee\*, “Monolithic assembly of elastic apatite with sub-nanometer scale integration of dual-phase organo-inorganic structures” Oral presentation, *ACS Fall 2025*, Washington DC, USA. August 17–21.
- [4] **C. H. Park**, J. W. Lee, B. Lee, D. Lee, J.-H. Lim, S.-W. Lee, P. R. Sundharbaabu, J. Bae, D. S. Yoon, E.-H. Lee, J. H. Heo, J. H. Lee, “DNA-Assembled Elastic Apatite Monoliths with Sub-Nanometer Precision”, Oral presentation, *The Korean Ceramic Society Spring 2025*, Yeosu, Korea. April 16–18.
- [3] **C. H. Park**, J. Bae, J. W. Lee, S. Chae, S. Oh, S. H. Kim, J.-Y. Choi, J. H. Lee\*, “Inorganic Molecular Nanowire-Based Extracellular Matrix-Mimicking Scaffolds for Tissue Engineering”, Oral presentation, *The 7th International Conference on Electronic Materials and Nanotechnology for Green Environment (ENGE 2022)*, Jeju, Korea. November 6–9.
- [2] **C. H. Park**, S. H. Jeong, J. W. Lee, M. J. Kim, J. H. Heo, J. H. Lee\*, “Biomolecular Adenosine Triphosphate as One Compacted Green Flame Retardant through Char formation with extreme intumescence”, Oral presentation, *The Korean Ceramic Society Fall 2021*, Jeju, Korea. November 3–5.
- [1] **C. H. Park**, C. Kim, J. W. Lee, J. H. Heo\*, D.-H. Kim, G. S. Yi, H. C. Kang, H. S. Jung\*, H. Shin\*, J. H. Lee\*, “Fabrication of Natural Bone Mimetic Hydroxyapatite-Based Scaffold Containing Nanopores for the Improvement of Bone Regeneration”, Oral Presentation, *The Korean Institute of Metals and Materials Spring 2021*, Hoengseong, Korea. April 28–30.

#### Poster Sessions

- [7] **C. H. Park**, J. W. Lee, B. Lee, D. Lee, J.-H. Lim, S.-W. Lee, P. R. Sundharbaabu, J. Bae, D. Shin, D. S. Yoon, E.-H. Lee, J. H. Heo\*, J. H. Lee\*, “Enhancing the Mechanical Properties of Apatite: A DNA-Driven Approach” Poster presentation (online), *Royal Society of Chemistry’s 2025 #RSCPoster Conference* on LinkedIn.
- [6] **C. H. Park**, J. W. Lee, B. Lee, D. Lee, J.-H. Lim, S.-W. Lee, P. R. Sundharbaabu, J. Bae, D. S. Yoon, E.-H. Lee, J. H. Heo, J. H. Lee, “DNA-Based Monolithic Assembly of Elastic Apatite with Dual Organo-Inorganic Structuring at the Sub-Nanometer Scale”, Poster presentation, *The Korean BioChip Society Spring 2024*, Jeju, Korea. November 13–15.
- [5] **C. H. Park**, J. W. Lee, B. Lee, D. Lee, J.-H. Lim, S.-W. Lee, P. R. Sundharbaabu, J. Bae, D. S. Yoon, E.-H. Lee, J. H. Heo, J. H. Lee, “Elastic DNAPatite with Sub-Nanometer Scale Nacre-Mimetic Structures”, Poster

presentation, *The Korean Ceramic Society Fall 2024*, Seoul, Korea. October 16–18.

- [4] **C. H. Park**, H. Song, S. H. Jeong, J. H. Heo\*, J. H. Lee\*, “Sustainable Cotton Fabric Treatment Using Adenosine Triphosphate dye for Enhancing Flame Retardancy and Color Retention”, Poster presentation, *The Korean BioChip Society Spring 2023*, Yeosu, Korea. May 31–June 2.
- [3] **C. H. Park**, J. Bae, J. W. Lee, S. Chae, S. Oh, S. H. Kim, J.-Y. Choi, J. H. Lee\*, “Molecular Chain Based on Inorganic Nanowire as Extracellular-Matrix Mimicking Scaffold with Bio-essential Properties”, Poster presentation, *The Korean BioChip Society Fall 2022*, Jeju, Korea. November 2–4.
- [2] **C. H. Park**, S. Kannappan, D.-H. Kim, S. H. Jeong, J. Bae, E. B. Kim, J. W. Lee\*, J. H. Lee\*, “Biomimetic Inorganic Molecular Nanowires as a Muscle Extracellular-Matrix Scaffold for Enhanced Bioactivity”, Poster presentation (online), *The Korean BioChip Society Fall 2021*, Korea. November 17–19.
- [1] **C. H. Park**, J. W. Lee, S. Chae, S. Oh, S. H. Kim, K. H. Choi, J. H. Heo, J.-Y. Choi\*, J. H. Lee\*, “Single-Chain Atomic Crystals as Extracellular Matrix-Mimicking Material with Exceptional Biocompatibility and Bioactivity”, Poster presentation, *The Korean Ceramic Society Fall 2020*, Daejeon, Korea. November 23–25.

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## SCHOLARSHIPS & AWARDS

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|-------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Best Oral Award</b> , International Conference on Nanogenerators and Piezotronics (NGPT 2026)                              | 2026         |
| <b>Academic Excellence Award</b> , 8 <sup>th</sup> Brain Korea 21 Outstanding Graduate Student Awards in AMSE, SKKU           | 2025         |
| <b>Best Oral Presentation Award</b> , Conference of The Korean Ceramic Society Spring (KCerS)                                 | 2025         |
| <b>Best Poster Presentation Award</b> , Conference of The Korean Ceramic Society Fall (KCerS)                                 | 2024         |
| <b>Best Presentation Award</b> , Research Matters Plus, SKKU                                                                  | 2023         |
| <b>Academic Excellence Scholarship</b> ,<br>- Half-tuition scholarship for the duration of the graduate program in AMSE, SKKU | 2020–2023    |
| <b>Brain Korea 21 Scholarship</b> , National Research Foundation of Korea (NRF)                                               | 2020–Present |
| <b>Samsung Dream Class Scholarship</b> , Samsung Welfare Foundation                                                           | 2019         |

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## PATENTS

- [6] J. H. Lee, J. H. Heo, S. H. Jeong, **C. H. Park**, H. Song, A Flame-Retardant Plastic Based on Adenosine Triphosphate and Epoxy Resin (South Korea – Application No. 10-2021-0101426; Registration No. 10-2559919).
- [5] J. H. Lee, J. H. Heo, **C. H. Park**, S. H. Jeong, H. Song, A Flame-Retardant Plastic Based on Adenosine Triphosphate and Polylactic Acid (South Korea – Application No. 10-2021-0101871; Registration No. 10-2632538).
- [4] J. H. Lee, J. H. Heo, **C. H. Park**, S. H. Jeong, H. Song, A Flame-Retardant Dye Using Adenosine Triphosphate as an Eco-Friendly Phosphorus-Based Flame-Retardant Additive (South Korea – Application No. 10-2021-0102362; Registration No. 10-2494785).
- [3] J. H. Lee, J. H. Heo, **C. H. Park**, S. H. Jeong, H. Song, A Method for Enhancing the Flame Retardancy of Polypropylene Resin with Adenosine Triphosphate (South Korea – Application No. 10-2021-0101463; Registration No. 10-2559916).
- [2] J. H. Lee, J. W. Lee, **C. H. Park**, J. H. Heo, T.-Y. Lee, D. H. Kim, An Organic-Inorganic Hybrid Monolithic Compound Using Nucleic Acid Monomers (ADP, AMP, TMP, GMP, CMP) (South Korea – Application No. 10-2021-0000787; Registration No. 10-2525363).
- [1] J. H. Lee, J. W. Lee, T.-Y. Lee, J. H. Heo, **C. H. Park**, D. H. Kim, An Organic-Inorganic Hybrid Monolithic Compound Using Phosphorylated Amino Acids (Phosphoserine, Phosphotyrosine, Phosphothreonine) (South Korea – Application No. 10-2021-0000751; Registration No. 10-2525360), Technology transferred (Recipient: CG Bio incorporation) Dec. 2022.

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## RESEARCH EXPERIENCE

**Graduate Student Researcher** | Biological & Nanoscale Materials Lab (BNML), SKKU

**Ceramic-supported Functional Nanomaterials**

Dec. 2024 – present

Co-principal investigator: Prof. Jung Heon Lee (SKKU) & Prof. Joohoon Kang (Yonsei Univ.).

- Contributed to the development of a material exhibiting high chemical resistance and catalytic performance through the sequential process of metal-catalyzed reaction and crystallization on a DNA scaffold.
- *Manuscript in preparation.*

### CHIRON: Ceramic Hybrid Interfaces foR Optimal cell stimulation

Dec. 2024 – present

Co-principal investigator: Prof. Jung Heon Lee (SKKU) & Prof. Tobias Fey (FAU).

Funded by the National Research Foundation of Korea (RS-2024-00438009)

- **The conceptual designing** of an international joint research program (Korea-Germany), **defining** key research objectives and **formulating** experimental strategies for developing advanced ceramic-hybrid interfaces for optimal cell stimulation.

### Development of High-Toughness Elasticeramic Materials

Mar. 2020 – Feb. 2025

Co-principal investigator: Prof. Jung Heon Lee (SKKU) & Dae Sung Yoon (Korea Univ.) & Eun-Ho Lee (SKKU)

Funded by the National Research Foundation of Korea (NRF-2020R1A2C2006100).

- Engineered a novel elasticeramic material with an organic-inorganic hybrid structure by developing a DNA-backbone synthesis method. **Characterized its unique mechanical properties using nanoindentation and UTM, and confirmed its structure via TEM and XRD**, leading to a co-first author publication in *Advanced Materials*.
- Publication: *Advanced Materials*, 36(41), 2406179 (2024).

### Development of Hydroxyapatite Based Artificial Bone Tissue Generation

Mar. 2020 – Dec. 2022

Principal investigator: Prof. Jung Heon Lee (SKKU)

Funded by the National Research Foundation of Korea (NRF-2018M3C1B7021997).

- **Led in vitro validation**, leading experiments with MC3T3-E1 pre-osteoblast cells to assess cytotoxicity, proliferation, and biocompatibility, directly.
- Publications: *Biomaterials Research*, 26(1), 7 (2022); *Journal of the Korean Ceramic Society*, 59(3) (2022).

### Functional Nanomaterials for Thermal Management & Flame Retardancy

Mar. 2021 – Jan. 2025

Principal investigator: Prof. Jung Heon Lee (SKKU)

- **Developed a portfolio** of sustainable flame retardants from biomolecules (ATP), creating multifunctional dyes, polymer coatings, and nanofillers that significantly enhanced fire safety and smoke suppression in cotton fabrics, commercial polymers, and batteries, respectively.
- Publications: *Chemical Engineering Journal*, 506, 159912 (2025); *Chemical Engineering Journal*, 497, 154406 (2024); *ACS Applied Polymer Materials*, 5(6) (2023); *Journal of Cleaner Production*, 368 (2022).

## TEACHING EXPERIENCE

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### Mentoring junior students in BNML

Davin Shin (PhD student)

Feb. 2024 – Aug. 2026

### Research mentor, Undergraduate Research Program (URP), SKKU

Mar. 2022 – Aug. 2022

- Mentored an undergraduate student project focused on the synthesis of biochar-hydrogel composites for water purification.
- Provided hands-on training and guidance on experimental procedures and data analysis.

### Teaching assistant

Mar. 2021 – Aug. 2022

- Instructed and mentored undergraduate students on experimental procedures and safe instrument operation.
- Graded lab reports and examinations, providing constructive feedback to enhance student learning.

## SKILLS

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### Materials characterization

- Microscopy: Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM)
- Spectroscopy: X-ray Diffraction (XRD), X-ray Photoelectron Spectroscopy (XPS), FT-IR, UV-Vis, Circular Dichroism (CD)
- Particle Analysis: Dynamic Light Scattering (DLS), Zeta Potential
- Mechanical Properties: Nanoindentation, Universal Testing Machine (UTM) for fracture mechanics analysis

#### **Biological assays & cell culture**

- Cell Lines: MC3T3-E1 (pre-osteoblast), L929 (fibroblast)
- Techniques: Aseptic cell culture and maintenance, cytotoxicity assays (e.g., MTT, WST), cell proliferation analysis, fluorescence microscopy (Live/Dead staining).

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#### **REFERRAL INFORMATION**

##### **1. Professor Jung Heon Lee (Advisor)**

- Vice Dean, College of Engineering, Sungkyunkwan University, Suwon, South Korea
- Professor, School of Advanced Materials Science & Engineering, SKKU, Suwon, South Korea

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##### **2. Professor Eun-Ho Lee**

- Associate professor, School of Mechanical Engineering, SKKU, Suwon, South Korea

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##### **3. Professor Joohoon Kang**

- Associate professor, Department of Chemical and Biomolecular Engineering, Yonsei University, Seoul, South Korea

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##### **4. Professor Jin Woong Lee**

- Assistant professor, Department of Pharmaceutical Engineering, Dankook University, Cheonan, South Korea

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